CampusPartner: An Assistive Technology for Mobility Impaired Pedestrians

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Abstract
CampusPartner is an assistive technology that we are developing to support people with mobility impairments in planning and view their walking routes in advance. By viewing the route in advance, users can see an overview and detailed information about it as well as turn-by-turn instructions. CampusPartner will be a mobile application that will integrate existing services to provide navigation functionality. Users will be able to create a profile, which will include information such as obstacles and road types they would like to avoid, as well as their favorite or most commonly used routes. Additionally, users will be able to correct missing or inaccurate information, such as the absence of stairs on the map or temporary obstacles.

Author Keywords
Navigation; mobile application; route planning; mobility; accessibility; open source; participatory design.

CCS Concepts
•**Human-centered computing** → **Accessibility technologies;** *Participatory design;* Open source software;

Introduction
Route-planning applications such as Google Maps and Apple Maps are used by 177.7 million unique users monthly [3]. However, these mapping applications are poorly suited

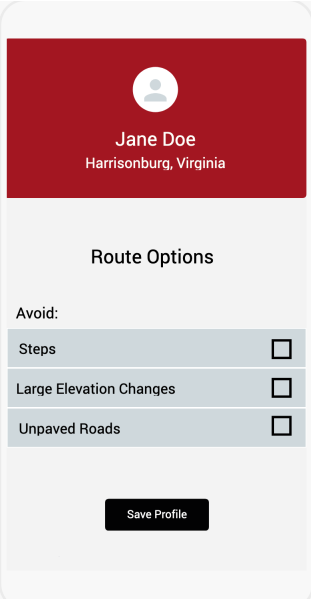

Figure 1: Profile creation mock-up in CP.

for those with mobility impairments (MIs). When creating a pedestrian route in Google Maps, there is not sufficient customization provided to the user in terms of obstacles they wish to avoid. These applications are not adequate for people with MIs [9]. Völkel and Weber provide the following definition for MIs, "Mobility impairments include all functional limitations which affect the ability of a person to reach a remote destination independently. A physical, cognitive, or sensory impairment or a combination of them may lead to a mobility impairment" [9, p.578]. Most widely used mobile applications are designed with vehicle use in mind and are not optimized for pedestrian navigation.

We are developing CampusPartner (CP) to assist MI pedestrians in planning their routes across campus and crowdsource map data updates. Accessibility issues are especially present on college campuses in the United States. CP is intended specifically for JMU, but has the potential to be extended beyond JMU and college campuses in future work. Often, people with MIs may need to plan ahead, researching everything from building entrances, the route to get there, and any other accommodations they may need [11]. CP would make this easier, because it would aggregate a lot of the information people would need to know all in one place and make it portable.

The HCI community focusing on cooperative and group work (such as ACM Group, CSCW, and SIGCHI more generally) has a long tradition of technologies applied to socio-technical problems such as physical access to facilities and information availability and accuracy in support of people with varying needs.

Review of Prior Work

Approaches to accessible routing have included applications such as AXSMap [4], which provides information about the accessibility of buildings using a rating system much like Yelp and AccessMap Seattle [2] which provides information such as elevation change, curb ramps, public elevators, etc. This approach is complemented by tools like Project Sidewalk which facilitates volunteer and paid crowdworkers in tagging accessibility issues found in Google Street View [8]. The open source RouteCheckr [10] and VirtualLeap & VirtualWalk [5] also inform our work on CP.

RouteCheckr (RC) aims to provide multi-criteria routing for MI pedestrians who are under-served by existing data sources. RC includes an annotation feature for updating geographical data with useful points of interest (POIs) and obstacles. CP will also faclitate users in improving (1) the data the application is using to operate and (2) the user experience.

VirtualLeap and VirtualWalk are two different forms of an application which uses turn-by-turn navigation and POIs to support the navigation of visually impaired users. The overall goal of both of these applications is to help visually impaired people to develop, "memorable representations of the real world" [5, p.281]. In the case of CP or other applications targeting university campuses, the use of POIs is particularly interesting because many of the places to which a user may need to navigate may not have street addresses or other disambiguating identifiers as in city building addresses.

Another similar application is Wheel Scout which determines barrier-free routes using information about temporary and static barriers [6]. However, there isn't any evidence of this project since 2014. Additionally, Wheelmap is a working example of mapping wheelchair accessible places but it doesn't include routing [7].

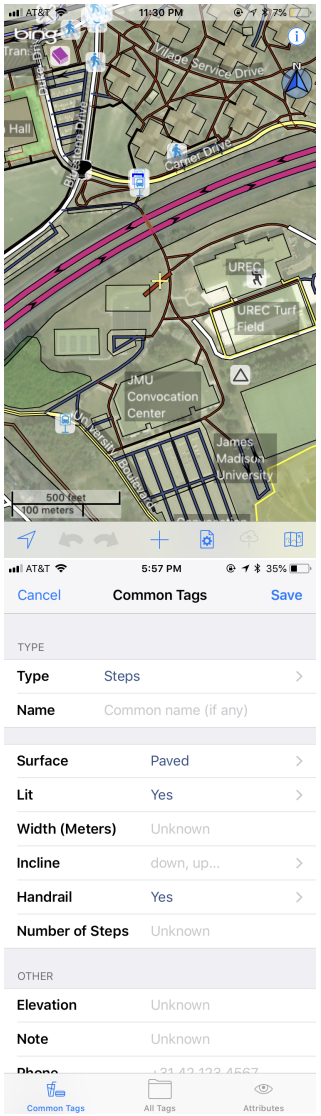

Figure 2: The map interface of the GoMap application and editing the data.

The Application

CP uses Open Street Maps (OSM) and Open Route Service (ORS). ORS provides an application programming interface (API) that facilitates the retrieval of directions and POIs. CP builds on earlier work (i.e. [5]) by (e.g.) including step-by-step navigation from the user-specified starting point and target destination(s). These navigational steps will be drawn from OSM with the user's settings, such as wheelchair accessibility, taken into consideration.

CP is designed primarily for those who are less familiar with the JMU campus. However, this app would also be beneficial for those who recently gained MIs and are unfamiliar with routes accessible to them. The first time a user opens the app, they will be prompted to log-in with their OSM credentials, this will help facilitate correcting missing data on the map, and create their profile. Their profile will primarily include any obstacles they would like to avoid. Once a user has logged in and created their profile (Figure 1), they are ready to start creating routes. They will be able to search for their source and destination and the route will be returned by ORS's directions endpoint. Once the route is retrieved, an overview will be shown to the user. This overview will be in the form of the route traced over the map tiles within the application.

CP is novel in its:

- focus on college campuses.
- crowdsourced, in situ map data corrections.
- aggregation of all pedestrian with MIs data sources.

Collaborative Emphasis

As this project is in its early stages, user studies remain future work. Once an initial prototype is completed, the app will be tested in collaboration with people with MIs. Beyond independence, *interdependent* relationships are needed to fully create access [1]. Particularly, in the case of navigation aids, putting the user in the subjugated place of simply following the instructions given to them by the assistive technology can be problematic. Campus Partner is designed to ally the user with the assistive technology and with past and future users to cooperatively find and verify the accuracy of the routes[1]. Developing assistive technologies must not just be developed for disabled people. The development process must also include them, as in the long history of Participatory Design methods. In addition to including the intended user base in the design process, CP will position users to contribute to the data source they are using within the app, improving it by using it and leaving space for users' sense of agency. Once a user encounters data that is inaccurate or finds that some data is missing, they can navigate to the GoMap application which utilizes the OSM API to add information and make changes to the map's data (Figure 2).

Conclusion

Although the ADA has made great strides to make sure that people with disabilities are not unfairly discriminated against, in many cases, the accessible entrances are hard to find, obstacles are not clearly labelled or marked, and historic buildings may be lacking in accessibility. Commonly used mapping applications do not provide the information and support needed by people with MIs. The need for way-finding applications that can support these needs are greatly needed. An application such as CP would alleviate some of the stress of visiting an unfamiliar place and not knowing whether you could reach your destination.

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